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Characterizing Speculative Decoding Dynamics for Large Language Models on Consumer-Class GPUs

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PART 1

The problem

Why consumer-GPU speculative decoding needs its own measurement

Decoding is sequential, and that is the bottleneck

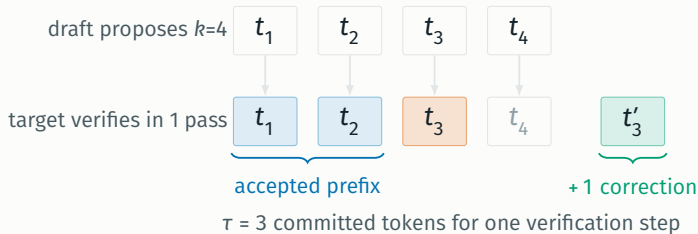
- Autoregressive decoding needs **one full forward pass of the target model per token**.
- Single-stream interactive latency is bound by *memory traffic*, not arithmetic.
- Speculative decoding is the standard fix — and is reported as a large, nearly free win.

Almost every published gain is measured on **data-centre** hardware. On a single RTX-class GPU, what actually happens?

OPEN DEPLOYMENT QUESTIONS

Which draft size?	0.5B? 1.5B?
Which proposal length?	$k=4$? 8? 16?
Greedy or sampling?	speed vs. quality
Does it transfer?	desktop → laptop

One speculative cycle



$$L = \frac{T_{\text{draft}} + T_{\text{verify}}}{\tau}$$

$$T_{\text{draft}} = k \cdot t_{\text{draft step}}$$

Numerator grows **linearly** in k ; τ saturates as acceptance falls. **So some k must lose.** Modified rejection sampling keeps the output distribution identical to the target's.

What is missing, and what we ask

THREE GAPS

- No cross-model *and* cross-hardware validation at the consumer tier.
- Speedups reported as bare means — no dispersion.
- Rarely a statistical test against a matched baseline.

A *characterization* paper, not a new decoding algorithm.

RESEARCH QUESTIONS

- RQ1** How much end-to-end speedup on consumer RTX hardware?
- RQ2** How do k and acceptance *jointly* set net latency?
- RQ3** How sensitive is the trade-off to deterministic vs. stochastic decoding?
- RQ4** Do rankings and gains transfer RTX4090 → RTX5090-laptop?

Experimental protocol

MODELS (same-family pairings)

Qwen2.5-3B-Instruct $\leftarrow$ 0.5B draft

Qwen3-4B-Instruct $\leftarrow$ 0.6B draft

FP16, matched tokenizers, KV cache both sides.

HARDWARE

	RTX4090	RTX5090-L
Bandwidth (GB/s)	1008	896
CUDA cores	16384	10496
Power (W)	450	≤ 150

Fixed-policy grid. $k \in \{4, 8, 16\} \times \{\text{det.}, \text{stoch.}\} = 6$ configurations *per family per device*, each against a **regime-matched** autoregressive baseline.

CONTROLS

Deterministic $T=0$, top- $p=1$; stochastic $T=0.7$, top- $p=0.9$.

1000 samples, frozen manifests: GSM8K 300, MMLU 500, CNN/DM 200.

Seed 42 primary; repeat seeds 123/999 at $k=4$.

PART 2

What we measured

Speedup, acceptance, token timing, and what transfers

RQ1 — on the RTX4090, it works

3.07_x

Qwen2.5 peak, deterministic
 $k=4$

2.85_x

Qwen3 peak, deterministic
 $k=4$

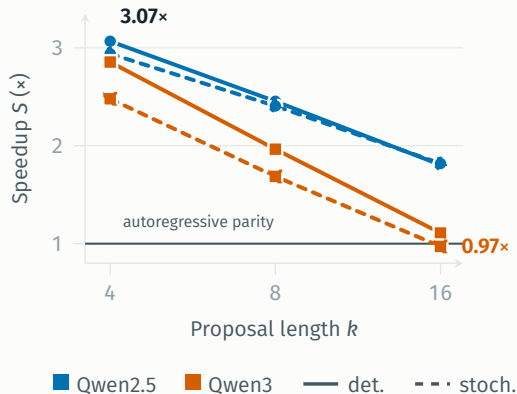
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Configurations above
baseline

	$k = 4$		$k = 8$		$k = 16$	
Speedup S	det	stoch	det	stoch	det	stoch
Qwen2.5 (3B $\leftarrow$ 0.5B)	3.0674	2.9394	2.4542	2.4063	1.8050	1.8185
Qwen3 (4B $\leftarrow$ 0.6B)	2.8548	2.4798	1.9648	1.6869	1.1091	0.9719

Both families peak at the *smallest* k we tested. Qwen3 stochastic $k=16$ is the single loss.

RQ2 — bigger blocks, slower generation



MEAN SPEEDUP, $k=4 \rightarrow k=16$

Qwen2.5 2.9164 $\rightarrow$ 1.8118

Qwen3 2.6673 $\rightarrow$ 1.0405

TOKEN TIMING DEGRADES TOO (Qwen3, RTX4090, det.)

TTFT 132.6 $\rightarrow$ 349.6 ms

TPOT 41.2 $\rightarrow$ 94.2 ms/tok

B_{eff} rises (1.33 $\rightarrow$ 1.96) while latency worsens. The overhead inflection arrives **before $k=8$** .

RQ3 — deterministic decoding is the safer default

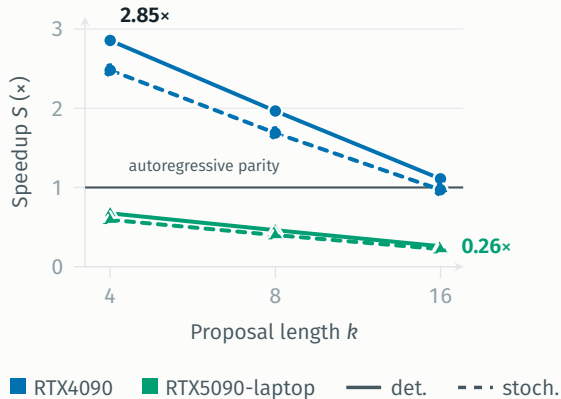
Qwen3, RTX4090	$k=4$	$k=8$	$k=16$
Deterministic S	2.8548	1.9648	1.1091
Stochastic S	2.4798	1.6869	0.9719
Advantage	+0.3750	+0.2779	+0.1372

Not universal. Qwen2.5 follows the same ordering at $k=4$ and $k=8$, then reverses slightly at $k=16$ — 1.8050 deterministic vs. 1.8185 stochastic.

- Greedy drafting agrees with greedy verification more often $\Rightarrow$ higher α (0.336 vs. 0.267 at $k=4$).
- The advantage *shrinks* with k : once acceptance is low, the regime hardly matters.

If you need sampling, expect to pay about **13% of your speedup** for it at low k .

RQ4 — same policy, different GPU, and it all inverts



0.67x

Best RTX5090-laptop configuration
(deterministic $k=4$) — all six span 0.22–0.67x

Every configuration is **slower than baseline** — yet acceptance equals or beats the RTX4090 at every k (0.335 vs. 0.336 at $k=4$; 0.180 vs. 0.129 at $k=16$).

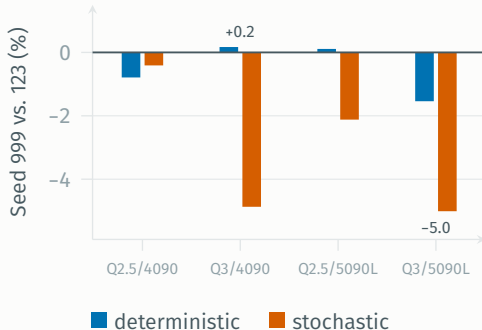
The loss is in per-step execution cost, not in the algorithm.

Is the effect real? Statistics and stability

PAIRED ONE-SIDED t -TESTS, $H_0: S \leq 1$
paired by sample, Holm–Bonferroni across 6 configs
(Qwen3, RTX4090, $n=1000$)

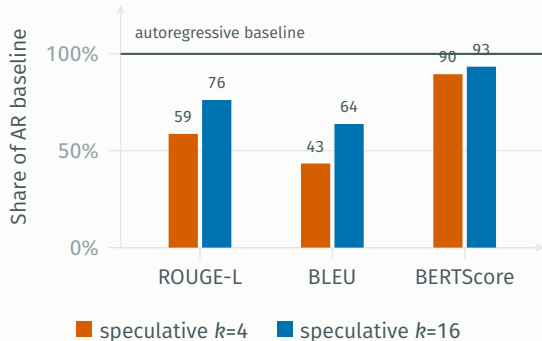
Config	d_S	p_S
det $k=4, 8, 16$	1.18 / 1.10 / 0.29	all $< 10^{-18}$
stoch $k=4, 8$	1.24 / 1.13	both $< 10^{-179}$
stoch $k=16$	-0.08	0.996

Dispersion is large. Per-sample S (det.) goes from 2.7561 ± 0.5950 at $k=4$ to 1.3549 ± 0.6533 at $k=16$ — at $k=16$ the standard deviation is half the mean.



Repeat seeds at $k=4$ only: deterministic within 1.5%, stochastic 0.4–5.0%. Regime ordering preserved everywhere.

Speed is not free: summarization quality



DETERMINISTIC, CNN/DAILYMAIL, $n=200$

	ROUGE-L	BLEU	BERTScore
AR baseline	19.58	3.20	0.793
Spec. $k=4$	11.50	1.39	0.710
Spec. $k=16$	14.92	2.04	0.741

The *fastest* configuration is also the *weakest* on summarization, and quality recovers as k grows — moving against the speed trend. A real frontier, not free acceleration.

PART 3

What it means

Bounded claims, and two rules you can deploy on

What we can and cannot claim

HYPOTHESES

- H1** $S > 1$ — **supported** on RTX4090 (11/12), **rejected** on RTX5090-laptop.
- H2** $k=8 > k=4$ — **rejected**; speedup falls monotonically.
- H3** B_{eff} up, latency worse — **supported**.
- H4** Deterministic faster — **mostly supported** (one reversal).

THREATS TO VALIDITY

- One PyTorch/Transformers stack; no extrapolation to other runtimes or GPUs.
- Same-family Qwen pairings only.
- No long-context, safety, or factual-consistency evaluation.
- Repeat seeds cover $k=4$ only.
- CUDA-graph capture differs across run families — reported as context, *not* as a measured cause.

Policy ranking is portable.
Absolute speedup is not.

Takeaways

1. **Use a short proposal.** $k=4$ dominates $k=8$ and $k=16$ on both GPUs and both families — the inflection is earlier than expected.
2. **Prefer deterministic decoding** at matched k ; sampling costs about 13% of the speedup at $k=4$.
3. **Benchmark on your target stack.** RTX4090 gave 3.07 $\times$; the same code on an RTX5090 laptop was *sub-baseline everywhere*, with acceptance equal or better.
4. **Report dispersion and quality, not just a mean speedup.** Per-sample spread is large; quality moves against the speed trend.

Future work: the full grid across seeds, broader hardware and model coverage, longer contexts, factuality-sensitive metrics.

PART 4

Backup

Full grids, acceptance portability, metrics, hardware

Backup — full Qwen3 RTX4090 grid

Config	S	α	B_{eff}	T_{mean} (s)	TTFT (ms)	TPOT (ms)
$k=4$, det	2.8548	0.3362	1.33	4.8881	132.57	41.23
$k=4$, stoch	2.4798	0.2672	1.06	5.6196	134.54	45.51
$k=8$, det	1.9648	0.2378	1.86	7.1024	204.99	58.76
$k=8$, stoch	1.6869	0.1916	1.50	8.2612	207.82	64.67
$k=16$, det	1.1091	0.1292	1.96	12.5817	349.58	94.24
$k=16$, stoch	0.9719	0.1082	1.64	14.3390	354.82	106.03

Target Qwen3-4B-Instruct; draft Qwen3-0.6B-Instruct; $n=1000$.

Backup — full Qwen2.5 RTX4090 grid

Config	S	α	B_{eff}	T_{mean} (s)	TTFT (ms)	TPOT (ms)
$k=4$, det	3.0674	0.4212	1.66	3.7210	102.35	29.06
$k=4$, stoch	2.9394	0.4080	1.61	3.9631	105.26	30.89
$k=8$, det	2.4542	0.3515	2.72	4.6508	162.76	36.73
$k=8$, stoch	2.4063	0.3444	2.67	4.8411	165.95	38.04
$k=16$, det	1.8050	0.2539	3.76	6.3234	277.43	49.34
$k=16$, stoch	1.8185	0.2516	3.73	6.4059	278.85	49.81

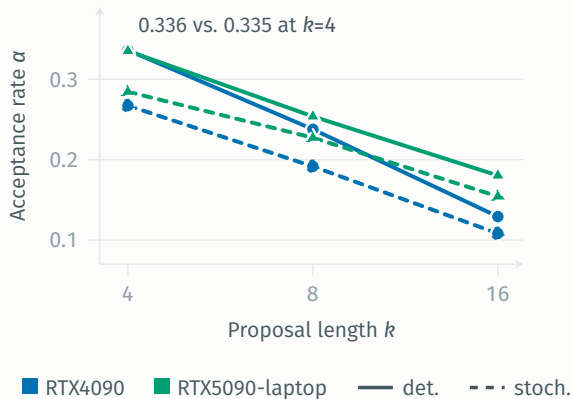
Target Qwen2.5-3B-Instruct; draft Qwen2.5-0.5B-Instruct; $n=1000$. Higher α and B_{eff} than Qwen3 at every k .

Backup — full Qwen3 RTX5090-laptop grid

Config	S	α	B_{eff}
$k=4$, det	0.6743	0.3353	1.32
$k=4$, stoch	0.5924	0.2845	1.06
$k=8$, det	0.4613	0.2538	1.85
$k=8$, stoch	0.3995	0.2273	1.50
$k=16$, det	0.2588	0.1803	1.96
$k=16$, stoch	0.2222	0.1543	1.64

Every configuration sits below parity, while α and B_{eff} match or exceed the RTX4090 at the same k .

Backup — acceptance portability



Qwen3 acceptance on the laptop is **identical** at $k=4$ (0.335 vs. 0.336) and **higher** at $k=8$ and $k=16$ (0.254 vs. 0.238; 0.180 vs. 0.129).

The speculative algorithm behaves at least as well there. The **runtime economics** are what differ.

Backup — success criteria and metrics

STATED SUCCESS CRITERIA

- Primary: mean $S \geq 1.3\times$ with Holm–Bonferroni-corrected $p_S < 0.05$.
- Primary: absolute quality drop ≤ 1.0 point on GSM8K and MMLU, deterministic regime.
- Secondary: best (k , regime) under quality constraints; dispersion; seed stability; cross-hardware ranking transfer.

KEY METRIC DEFINITIONS

- S = baseline latency / speculative latency
- α = accepted / proposed draft tokens
- B_{eff} = accepted draft tokens / verify steps
- TTFT = first token time – request start
- TPOT = (completion – first token) / output tokens
- d_S = Cohen's d , paired baseline vs. speculative latency

Backup — hardware profiles

Component	RTX4090 desktop	RTX5090 laptop
GPU memory	24 GB GDDR6X	24 GB GDDR7
Memory bandwidth	1008 GB/s	896 GB/s
CUDA cores	16384	10496
Architecture	Ada Lovelace	Blackwell
Power profile	450 W TGP class	up to 150 W TGP (OEM-dependent)
Software stack	PyTorch + Transformers, pinned per run family	

The laptop part is the *newer* architecture but has lower bandwidth and roughly **one third of the power budget**.